Complexity of Jet Reconstruction Algorithms

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UO
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  - Measuring Algorithmic Complexity
  - Analysis of FastJet, SISCone, and their predecessors

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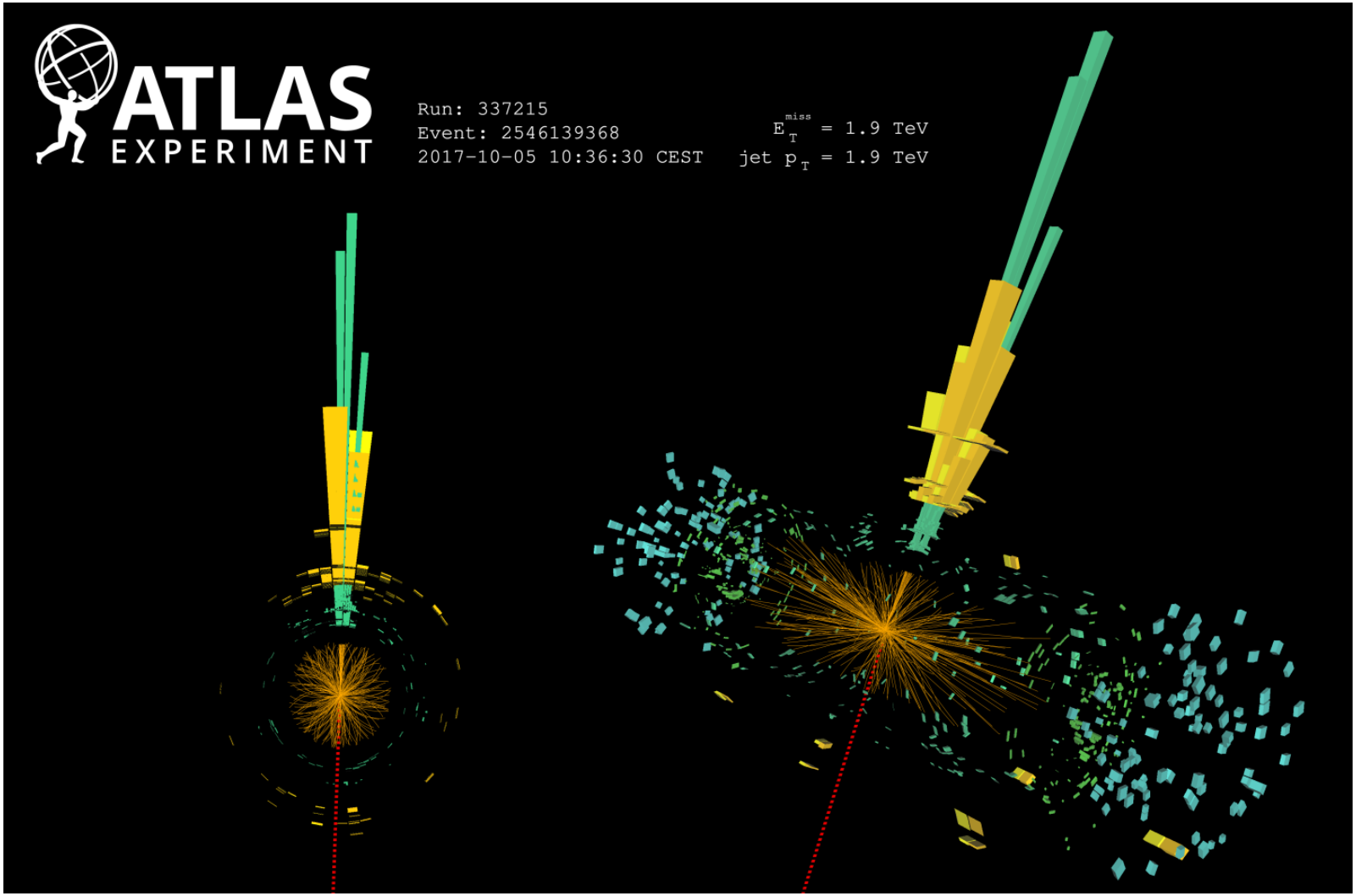
- Jets:
  - A collimated spray of stable particles arising from fragmentation and hadronization of a parton after a collision.

Review of Jet Algorithms (cont.)

Example Jet



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  - Necessary for faithful comparison of data to experiment

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  - Examples include K_t, $\bar{K}_t$, and Cambridge/Aachen

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  - Also, $o()$, $\Theta(x)$, $\Omega()$, $\omega()$ with related definitions

Complexity of K_t (pre-FastJet)

The Naïve Algorithm

Algorithm Naïve K_t

```
repeat
    for particle pair (i, j) do
        d_Bi ← p_ti^2                          ▷ Calculated once for each i
        R_ij^2 ← (η_i − η_j)^2 + (φ_i − φ_j)^2
        d_ij ← min(p_ti^2, p_tj^2) R_ij
    end for
    d_min ← min({d_ij} ∪ {d_Bi})
    if d_min is d_Bi then
        Add particle i to list of jets, remove from list of particles
    else
        Merge particles i, j
    end if
until no particles remain
```

Pre-FastJet K_t Complexity (cont.)

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$$\mathcal{O}(N + N^2 + N(1 + N + N^2 + 1)) = \mathcal{O}(N^3)$$

- Also easy to see that the space complexity is $\mathcal{O}(N^2)$.

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- That is, particles i and j are geometric *nearest neighbors*
- Therefore, we don't need the $\mathcal{O}(N^2)$ table d_{ij}!
- We only need an n-element list of nearest neighbors $d_{i\mathcal{G}_i}$

Complexity of FastJet

The Smart Algorithm

Algorithm `FastJet` K_t

```
for particle i do
    G_i ← nearest neighbor of particle i
    d_iG_i, d_Bi calculated as previously
end for
repeat
    d_min ← min({G_i} ∪ {d_Bi})
    Merge or Remove according to d_min as previously
    Update G_i, d_iG_i, d_Bi.
until no particles remain
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    - Updating $\mathcal{G}_i$ in $\mathcal{O}(N)$ is a nontrivial geometric affect
- However, we can do better!

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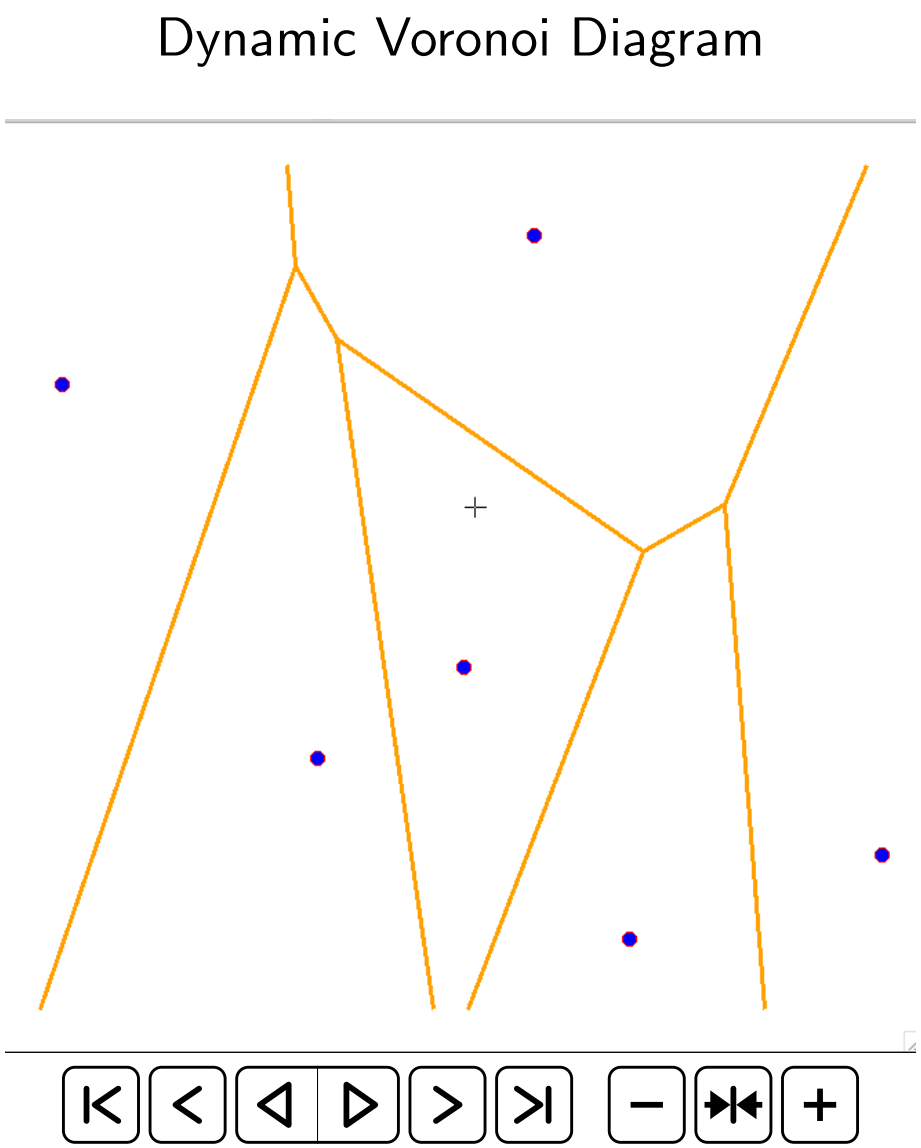

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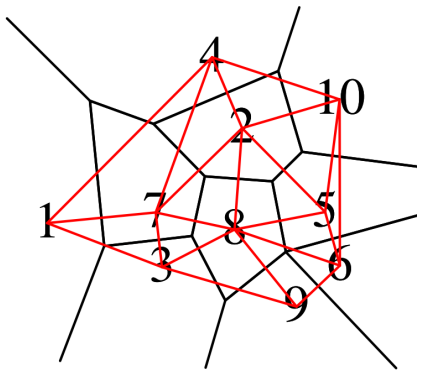
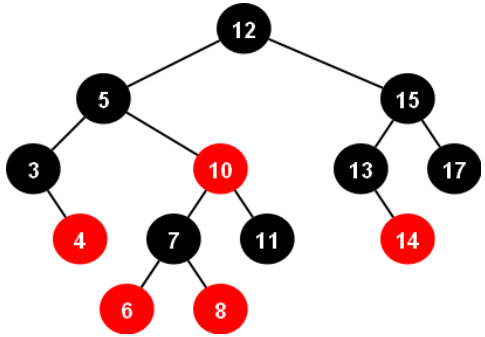
Better Complexity Analysis

- A Voronoi diagram for NN calculations





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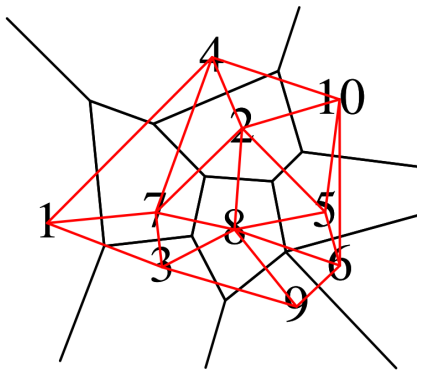

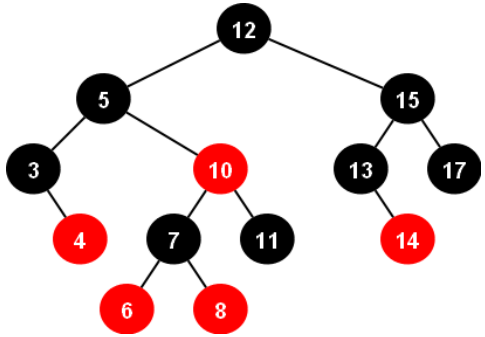

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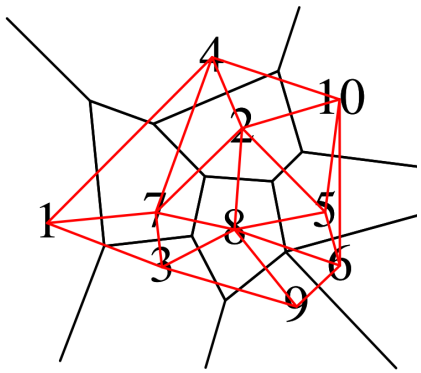

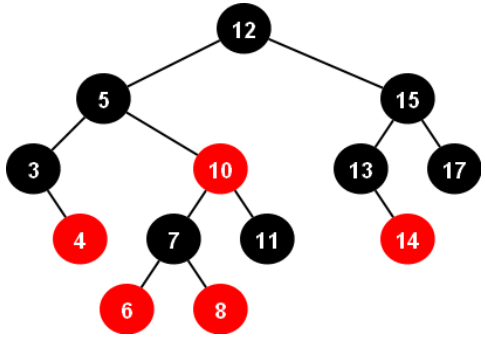

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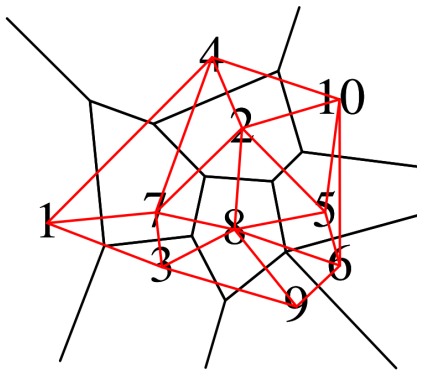

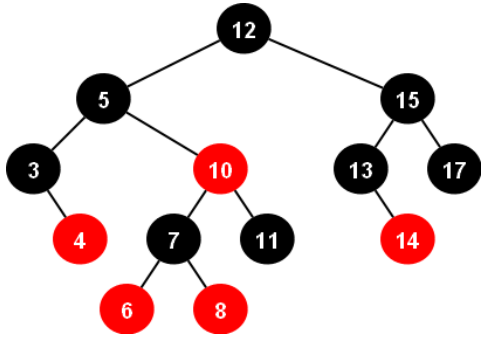

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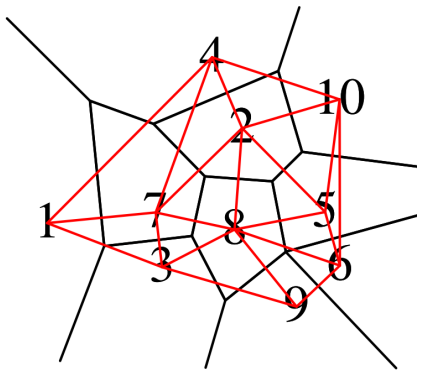

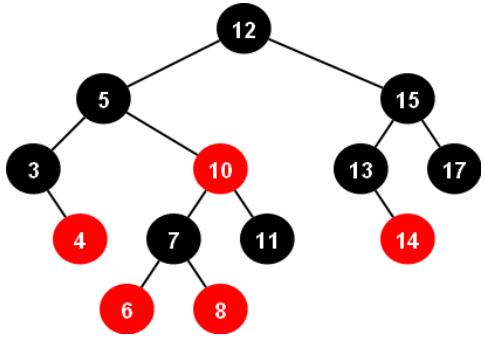

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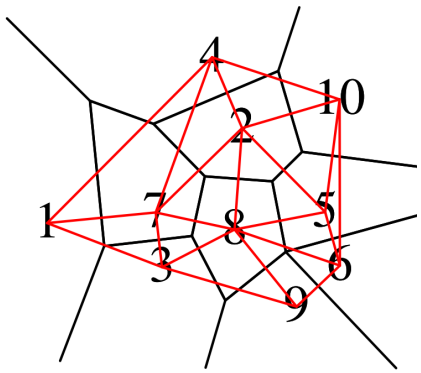

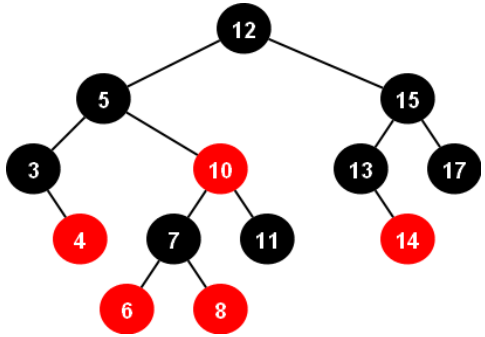

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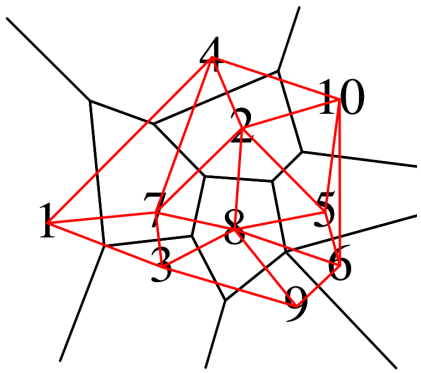

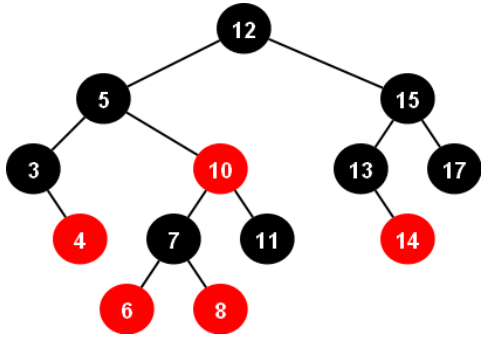

Complexity of FastJet (cont.)

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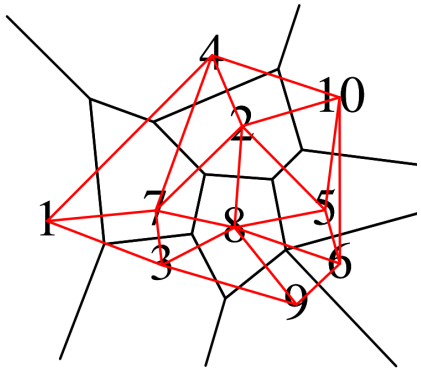

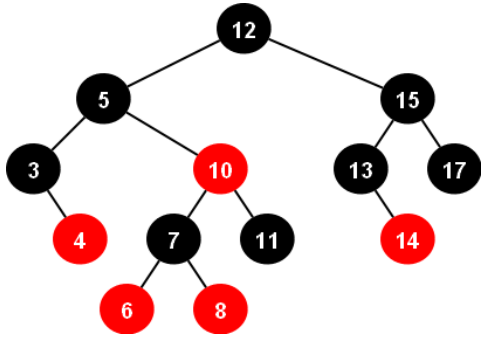

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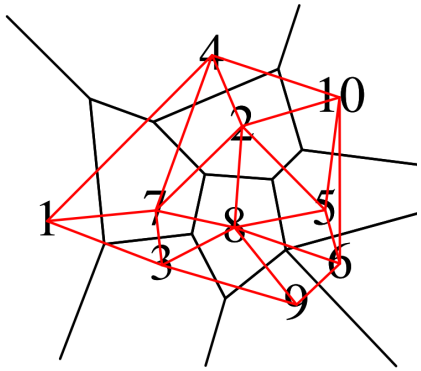

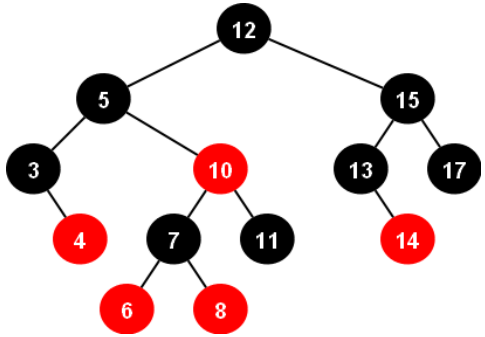

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Complexity of IC-SM

A Naïve Cone-Finding Algorithm

Algorithm Iterative Cone with Split-Merge

```
repeat
    p_t^* ← max({p_ti})                                   ▷ seed axis
    repeat
        p_j ← sum of momenta of particles within R of seed axis
        if p_t^* == p_j then
            Label p_t^* a protojet, remove all particles
        else
            p_t^* ← p_j
        end if
    until The jet axis and seed axis coincide
until No seeds remain
Run Split-Merge on the protojets
```

Complexity of IC-SM

The Split-Merge Procedure

Algorithm Split-Merge

```
Remove all protojets with p_t < p_{t,cut}
repeat
    Determine hardest protojet i
    Determine hardest protojet j such that i and j share particles
    if no such j exists then
        i is a final jet; remove particles
    else
        if p_{t,shared} < f × p_{tj} then
            Split particles between protojets
        else
            Replace i and j with their merger
        end if
    end if
until no protojets left
```

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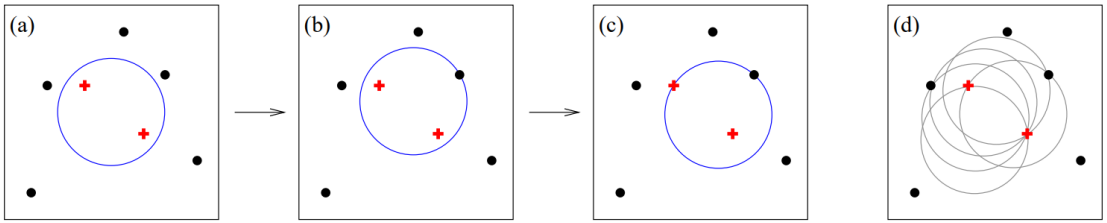
  - Merging takes $\mathcal{O}(Nn)$
- Total complexity: $\mathcal{O}(N^2 n)$ (can be optimized to $\mathcal{O}(N^2)$ with 2D tree data structures)

Complexity of SISCone

Improved Cone-Finding



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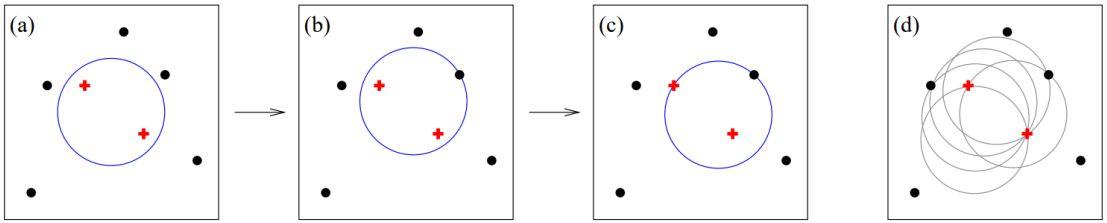

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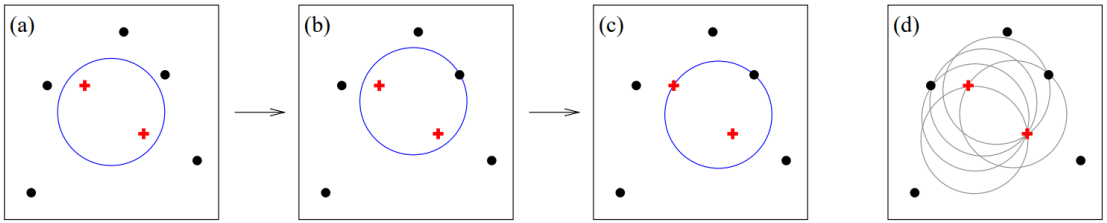

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  - (particularly by those $N2^N$ codes...)

References

- Ryan Atkin 2015 J. Phys.: Conf. Ser. 645 012008
- G. P. Salam and G. Soyez, JHEP 0705 (2007) 086 [arXiv:0704.0292 [hep-ph]].
- M. Cacciari and G. P. Salam, arXiv:hep-ph/0512210.
- S. Fortune, in Proceedings of the second annual symposium on Computational geometry, p. 312 (1986).